

BNPL

Smoothing or Debt Trap *and for whom?*

THEORY OF CONSUMPTION ECONOMICS

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So what is BNPL?

Service lets a consumer to **split** a purchase into **few small installments** paid over a **short period**.

1

**You buy
\$100 item**

Online or in-store



2

**Pay \$25
at checkout**

First installment



3

**Three more
\$25 payments**

Every 2 weeks



4

**No interest
if on time**

Late fees if not

Overview

\$500B

projected global transaction
value by 2030

14%

of U.S. adults used BNPL in
2023 (up from 10% in 2021)

180M

U.S. loans in 2021, up from
16.8M just two years earlier

Theoretical Framework



Permanent Income Hypothesis

Friedman (1957)

Credit lets households smooth income. More access raises welfare.



Liquidity Constraints

Kaplan, Violante & Weidner (2014)

A household is liquidity constrained when it cannot borrow freely.



Behavioral Finance

Laibson (1997); Thaler (1985)

Installments feel cheaper, so people spend more than smoothing alone justifies.

Two opposite camps

SMOOTHING

BNPL helps constrained households

Di Maggio et al. (2022)

Constrained households use BNPL to smooth spending across pay cycles

Papich (2022)

Access lowers distress and credit scores rise modestly

Wheat et al. (2026)

A “liquidity valve” — used most when budgets are tight

OVERBORROWING

BNPL adds to financial strain

deHaan et al. (2024)

Overdraft and credit-card fees rise after adoption (10.6M consumers)

Maesen & Ang (2025)

Installments make people buy more often

Guttman-Kenney et al. (2023)

Many repay BNPL with credit cards (debt on debt)

KEY IDEA

“constrained” households



Wealthy hand-to-mouth

- Own assets (home, retirement) but little spare cash.
- Borrow to bridge a short cash-flow gap.

BNPL likely smooths → welfare ↑



Poor hand-to-mouth

- Little wealth of any kind.
- Borrow to replace income they will not get back.

BNPL risks lasting debt → welfare ↓

Both are labeled “liquidity constrained” **but their welfare outcomes point in opposite directions.**

The average hides who BNPL is for

BNPL users are not representatives of the population.

If the product **helps** stable households but **harms** fragile ones -> single average effect can point either way depending only on the **sample mix**.

Questions:

~~“does BNPL help or harm?”~~

but

“for whom?”

2×

Adults “just getting by” are about **twice as likely to use** BNPL as those who feel comfortable

1 in 5

Among adults who could not cover a **\$500** emergency, roughly one in five used BNPL in 2023

Source: Larrimore et al. (2024), Federal Reserve FEDS Notes.

RESEARCH DESIGN

Why a standard difference-in-differences isn't enough?

1

Selection bias

Adopters were already drifting into financial stress before adopting (Larrimore et al., 2024). A simple before-and-after comparison would blame BNPL for a decline already happen.

FIX An event study shows the pre-adoption years explicitly, so pre-trends can be checked.

2

One pooled effect hides groups

A single average can blend a benefit for one group with harm for another, leaving the sign to depend on the sample mix.

FIX An interaction by baseline liquidity separates the groups.

→ *A long-panel event study with an interaction by baseline liquidity.*

RESEARCH DESIGN

An event study with a heterogeneity term

$$Y_{it} = \alpha_i + \lambda_t + \sum \beta_k D_{i,t,k} + \sum \delta_k (D_{i,t,k} \times \text{Constrained}_i) + X'\gamma + \varepsilon$$

β_k

The path over time

How distress changes each year before and after adoption.

δ_k

The contribution

How constrained households differ from others at the same point after adoption.

α_i, λ_t

The controls

Household and year fixed effects strip out fixed traits and economy-wide shocks.

The δ_k terms are the heart of the proposal — the test the literature has not yet run cleanly.

DATA & IDENTIFICATION

DATA

PSID

Long household panel with detailed balance sheets and financial-stress measures.

SHED

Federal Reserve survey with direct BNPL-use questions since 2021.

Linking

Imputation and nearest-neighbor matching; agreement between them is a robustness check.

IDENTIFICATION

Pre-trend tests

Flat pre-adoption coefficients support a causal reading.

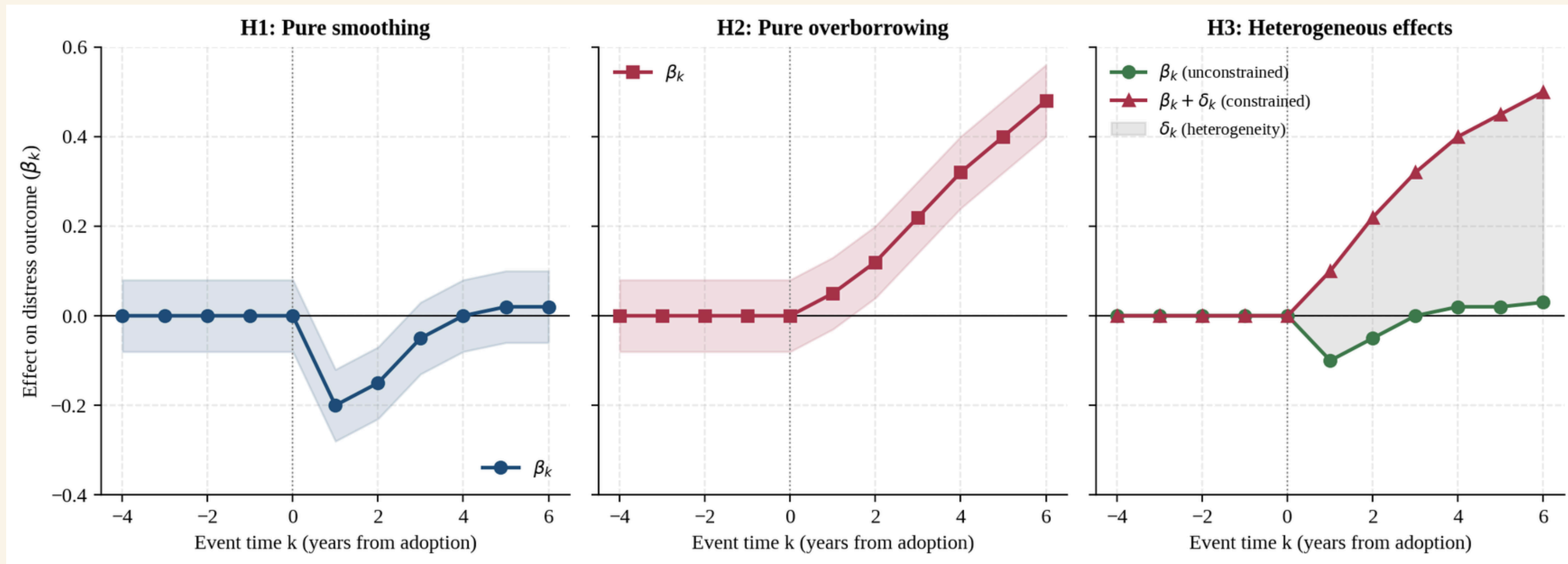
Robust estimators

Callaway & Sant'Anna (2021) and Sun & Abraham (2021) avoid staggered-DiD bias.

Scope

Identifies effects by group, not the average causal effect without an instrument or RCT.

HYPOTHESES



H1 Pure smoothing

$\beta_k < 0$ briefly, then returns to zero

H2 Pure overborrowing

β_k rises monotonically and persists

H3 Heterogeneous

$\delta_k > 0$: constrained households diverge

CONTRIBUTION & POLICY

The goal is not to call BNPL good or bad, but to learn which households it smooths and which it pushes into distress.

What should regulators do?



Consistent credit reporting

End the invisible-debt problem so lenders and consumers can see total BNPL exposure.



Cap loan stacking

- 63% hold multiple loans
- 33% borrow across providers
- No one seeing the total)



Minimum underwriting

Basic repayment checks at checkout, where today there are almost none.

Keep the smoothing function while limiting the debt-trap channel.

Thankyou